

Advanced EDA Interview Questions & Answers

A Comprehensive Reference Guide

Contents

1. Advanced EDA Fundamentals	1
2. Descriptive Statistics	3
3. Distribution Analysis	5
4. Missing Data	6
5. Outlier Analysis	8
6. Correlation and Relationship Analysis	9
7. Categorical Data EDA	11

1. Advanced EDA Fundamentals

Q: What is Exploratory Data Analysis (EDA), and how does advanced EDA differ from basic EDA? EDA is the process of investigating a dataset to summarize its main characteristics, uncover patterns, spot anomalies, and test assumptions, usually using visual and statistical methods. Basic EDA covers summary statistics and simple plots (mean, histograms, bar charts). Advanced EDA goes further: it examines distributional shape (skew, kurtosis, multimodality), missing-data mechanisms, multicollinearity, interaction effects, contextual/business-driven outlier reasoning, and segment-level or conditional analysis (e.g., relationships that change across subgroups).

Q: Why should EDA be performed before applying machine learning algorithms? Models assume things about the data (linearity, normality, independence, no leakage) that may not hold. EDA reveals data quality issues, feature relationships, and target leakage before they silently corrupt model training, helping you choose appropriate algorithms, transformations, and validation strategies rather than fitting a model blindly.

Q: What are the major objectives of EDA in a real-world data science project? Understand the structure and quality of the data; detect errors, missing values, and

outliers; understand distributions and relationships between variables; generate hypotheses; guide feature engineering; and validate whether the data actually supports the business question being asked.

Q: Explain the difference between univariate, bivariate, and multivariate analysis with examples. Univariate analysis examines one variable in isolation (e.g., a histogram of “age”). Bivariate analysis examines the relationship between two variables (e.g., a scatter plot of “age” vs. “income”). Multivariate analysis examines relationships among three or more variables simultaneously (e.g., a pair plot or a regression of “income” on “age,” “education,” and “region” together).

Q: What is the difference between EDA and data preprocessing? EDA is diagnostic and exploratory — it is about understanding the data. Preprocessing is corrective and preparatory — it is about transforming the data (imputing missing values, encoding categories, scaling) so it is ready for modeling. In practice EDA informs which preprocessing steps are needed, and the two are often interleaved.

Q: Why is understanding the business context important before performing EDA? Without business context, you can’t tell whether a value is an error or a legitimate extreme case, which variables are actually meaningful, or which relationships matter. Business context tells you what a “normal” range looks like, what the target variable really represents, and which findings are actionable versus statistical noise.

Q: What problems can occur if you perform statistical analysis without understanding the meaning of each feature? You risk misinterpreting codes as continuous numbers (e.g., a “zip code” column treated numerically), missing that a feature is actually a proxy for the target (leakage), drawing incorrect causal conclusions, mishandling units (e.g., mixing currencies), and applying inappropriate transformations to variables that are categorical in disguise.

Q: How do you determine whether a variable should be treated as numerical, categorical, ordinal, or datetime? Look at the nature of the values and how they’re used: numerical if arithmetic operations (mean, sum) are meaningful; categorical (nominal) if values are labels with no order (e.g., colors); ordinal if categories have a meaningful order but unequal/undefined spacing (e.g., satisfaction ratings); datetime if values represent points in time and need date arithmetic. Also check the number of unique values, cardinality, and how the variable is described in the data dictionary.

Q: What is the difference between data profiling and EDA? Data profiling is largely automated and descriptive: it produces statistics like counts, data types, unique values, and missingness per column, often via a tool (e.g., pandas-profiling). EDA is a broader, more manual and hypothesis-driven investigation that uses profiling as a starting point but adds visualization, relationship analysis, and business reasoning.

Q: What is meant by data quality assessment during EDA? It is the process of checking whether the data is accurate, complete, consistent, and reliable enough to be used — examining missing values, duplicate records, inconsistent formatting, invalid values (e.g., negative ages), and mismatches between related fields.

Q: Which aspects of data quality should be examined during EDA? Completeness (missing values), consistency (uniform formats/units), accuracy (values match

reality), validity (values within expected ranges/domains), uniqueness (duplicate records), and timeliness (data is current enough for the task).

Q: How can EDA help identify problems in a data collection process? Patterns such as suspicious spikes at round numbers, identical timestamps across many records, systematic missingness for a subgroup, or values that cluster at sensor/system limits often point to how the data was collected (e.g., a form defaulting to “0,” a sensor malfunctioning, or a survey question being skipped by certain respondents).

Q: Why is it dangerous to remove observations simply because they appear unusual? Unusual observations can be genuine and important — fraud cases, high-value customers, rare disease presentations — and removing them can bias the model and hide the very signal you care about. Blanket removal without investigation can also silently reduce sample size and distort the target distribution.

Q: How do you distinguish between a data error and a genuine extreme observation? Check whether the value is physically/logically possible (e.g., a negative age is an error; a 110-year-old person may not be), cross-validate against other fields and external context, check if similar extreme values recur, and consult domain experts or data owners when uncertain.

Q: What is the difference between an observation, feature, target variable, and identifier? An observation (row) is a single record/instance. A feature (column) is an attribute describing observations, used as model input. The target variable is the outcome you are trying to predict or explain. An identifier (e.g., customer ID) uniquely labels a row but carries no predictive information and should generally be excluded from modeling.

2. Descriptive Statistics

Q: Explain the difference between mean, median, and mode. When is each more useful? The mean is the arithmetic average, useful for symmetric distributions without extreme values. The median is the middle value when sorted, more robust for skewed data or data with outliers. The mode is the most frequent value, most useful for categorical data or to describe the most common value in a multimodal distribution.

Q: Why can the mean be misleading when a dataset contains extreme values? Because every value contributes equally to the mean’s calculation, a few very large or very small values can pull it far from where most of the data actually lies, giving a distorted sense of the “typical” value.

Q: When would you prefer median over mean? When the data is skewed or contains outliers — for example, household income, house prices, or hospital stay durations — since the median better reflects the “typical” observation.

Q: What is variance, and what does it tell you about a dataset? Variance is the average of the squared deviations from the mean. It quantifies how spread out the data points are; higher variance means values are more dispersed around the mean.

Q: Explain standard deviation intuitively. Standard deviation is the square root of variance, expressed in the same units as the original data, making it easier to interpret. It represents roughly how far, on average, individual data points deviate from the mean.

Q: What is the difference between range, variance, standard deviation, and IQR? Range is the difference between max and min (sensitive to outliers). Variance and standard deviation measure average squared/absolute spread around the mean (also outlier-sensitive). IQR is the range of the middle 50% of the data (Q3-Q1), robust to outliers and better for skewed distributions.

Q: Why is IQR often preferred over standard deviation for skewed data? Standard deviation is influenced heavily by extreme values because it squares deviations. IQR only considers the middle 50% of data, so it isn't distorted by long tails or a handful of extreme outliers, making it more representative of "typical" spread in skewed data.

Q: What does the coefficient of variation tell us? The coefficient of variation (standard deviation divided by the mean, often as a percentage) measures relative variability, allowing comparison of dispersion across variables or datasets with different units or scales.

Q: How can skewness be identified using descriptive statistics? By comparing the mean, median, and mode (mean > median suggests right skew, mean < median suggests left skew), computing the skewness coefficient directly, or visually inspecting a histogram or KDE plot for asymmetry.

Q: What does kurtosis tell you about a distribution? Kurtosis measures the "tailedness" of a distribution relative to a normal distribution. High kurtosis (leptokurtic) indicates heavier tails and more outliers; low kurtosis (platykurtic) indicates lighter tails and fewer extreme values.

Q: What is the difference between positive skewness and negative skewness? Positive (right) skewness means the tail extends toward higher values, with mean > median. Negative (left) skewness means the tail extends toward lower values, with mean < median.

Q: A dataset has a mean of 100 and a median of 70. What might this indicate about the distribution? The mean being much larger than the median suggests strong right (positive) skewness — a small number of very large values are pulling the mean upward while most observations cluster closer to 70.

Q: A dataset has mean = median = mode. What can you infer about its distribution? This is characteristic of a symmetric, unimodal distribution (like a perfect normal distribution), though it doesn't guarantee normality — other checks (skewness, kurtosis, Q-Q plots) are still needed.

Q: Why should you examine quantiles and percentiles during EDA? They reveal the shape and spread of the data beyond a single summary number, help detect skewness and outliers, and are useful for defining thresholds (e.g., top 10% of customers) or robust scaling methods.

Q: How would you explain the difference between the 25th percentile and

the 75th percentile to a non-technical person? The 25th percentile is the value below which a quarter of your data falls — the “low end” of the typical range. The 75th percentile is the value below which three-quarters of your data falls — the “high end” of the typical range. Together they bracket the middle 50% of your observations.

3. Distribution Analysis

Q: What is a probability distribution, and why is understanding distribution important during EDA? A probability distribution describes how values of a variable are spread across possible outcomes. Understanding it is important because it informs which statistical tests are valid, what transformations may help, how to detect outliers, and what modeling assumptions are reasonable.

Q: How can you determine whether a numerical feature approximately follows a normal distribution? Visually via histograms, KDE plots, or Q-Q plots, and statistically via tests like Shapiro-Wilk, Kolmogorov-Smirnov, or Anderson-Darling, along with checking skewness and kurtosis values close to zero.

Q: What is the importance of a histogram in EDA? Histograms show the frequency distribution of a numerical variable, revealing shape, central tendency, spread, skewness, and the presence of multiple modes or gaps at a glance.

Q: What information does a KDE plot provide? A Kernel Density Estimate plot provides a smoothed, continuous estimate of the underlying probability density, making it easier to see the overall shape of a distribution without the binning artifacts of a histogram.

Q: What are the limitations of relying only on histograms? Their appearance is sensitive to bin width and bin edges, which can create misleading impressions of the shape (hiding or exaggerating modes), and they don’t show fine details as smoothly as KDE plots.

Q: What is a Q-Q plot, and how can it help during EDA? A Quantile-Quantile plot compares the quantiles of your data against the quantiles of a theoretical distribution (usually normal). If points fall along the reference line, the data approximates that distribution; deviations reveal skewness, heavy tails, or other departures from normality.

Q: How would you identify whether a variable is right-skewed? Mean noticeably greater than median, a histogram/KDE with a long tail extending to the right, positive skewness coefficient, and a Q-Q plot curving away from the line at the upper end.

Q: What is a bimodal distribution, and what could cause it? A bimodal distribution has two distinct peaks. It can arise from a mixture of two underlying subpopulations (e.g., combining data from two customer segments, two machines, or two time periods) that each have their own typical value.

Q: What does a multimodal distribution suggest about the underlying data? It suggests the data may be a mixture of several distinct subgroups or processes rather than a single homogeneous population, and it may be worth segmenting the analysis by the underlying grouping variable.

Q: Suppose customer income has two peaks. What possible business explanations could exist? Different customer segments (e.g., individual vs. business accounts), different geographic regions with different cost-of-living/income levels, different acquisition channels, or a mix of full-time vs. part-time customers.

Q: What is the difference between a normal distribution and a log-normal distribution? A normal distribution is symmetric and bell-shaped on its original scale. A log-normal distribution is right-skewed on its original scale, but becomes approximately normal after taking the logarithm of the values — common for quantities that are products of many small multiplicative effects (e.g., income, stock prices).

Q: Why might income, transaction amount, or population variables be heavily right-skewed? Because these quantities are naturally bounded below by zero but have no natural upper bound, and a small number of very large values (wealthy individuals, large transactions, big cities) can pull the distribution's tail far to the right.

Q: When might a log transformation be useful during EDA? When a variable is heavily right-skewed and multiplicative in nature; a log transform compresses large values, makes the distribution more symmetric/normal-like, stabilizes variance, and can make relationships with other variables more linear.

Q: What problems can highly skewed variables create for downstream analysis? They can violate normality assumptions in statistical tests, distort distance-based algorithms (like k-means or KNN), inflate the influence of extreme values on models sensitive to scale (like linear regression), and make visualizations hard to interpret.

Q: How would you analyze a variable containing many zero values and a small number of very large values? Separate the analysis into two parts: the proportion of zeros (often modeled as a separate binary indicator) and the distribution of the non-zero values (often log-transformed). This “zero-inflated” pattern is common in spending, insurance claims, or usage data.

4. Missing Data

Q: What are the different types of missing data? MCAR (Missing Completely at Random), MAR (Missing at Random), and MNAR (Missing Not at Random).

Q: Explain MCAR, MAR, and MNAR. MCAR: missingness is unrelated to any observed or unobserved data — purely random (e.g., a random sensor glitch). MAR: missingness depends on other observed variables but not on the missing value itself (e.g., younger respondents more often skip an income question). MNAR: missingness depends on the value that is missing itself (e.g., people with very high income choose not to disclose it).

Q: Why is simply replacing every missing value with the mean often a poor strategy? It reduces variance, distorts the distribution's shape, weakens correlations with other variables, and can mask patterns in missingness (especially under MAR/MNAR) that carry real information.

Q: When is median imputation preferable to mean imputation? When the variable is skewed or contains outliers, since the median is more robust and better represents a “typical” value in that situation.

Q: When can mode imputation be used? For categorical variables, where mean/median don’t apply — imputing with the most frequently occurring category.

Q: What information can a missing-value heatmap provide? It visually shows which rows and columns have missing values and whether missingness clusters together (e.g., several columns missing for the same rows), hinting at systematic patterns or shared root causes.

Q: How can missingness itself become an important feature? Creating a binary “is_missing” indicator for a variable can capture predictive signal when the fact that a value is missing correlates with the outcome (e.g., customers who don’t provide a phone number may behave differently).

Q: Suppose customers with high income are more likely to leave the income field blank. What type of missingness might this represent? MNAR (Missing Not at Random) — the probability of missingness depends on the value itself (income), not just on other observed variables.

Q: A dataset contains 60% missing values in one feature. What factors would you consider before removing the feature? Whether the feature is critical to the business question or model, whether missingness itself is informative, whether the missing values can be reasonably imputed or reconstructed from other sources, and what would be lost if a large chunk of the data (or the feature entirely) were dropped.

Q: How would you investigate whether missing values are randomly distributed? Compare the distributions of other variables between rows with missing values and rows without, use statistical tests (e.g., Little’s MCAR test), and visualize missingness patterns using tools like a missingness heatmap or matrix plot.

Q: How can correlation between missingness indicators reveal data collection problems? If binary “is_missing” flags for multiple columns are highly correlated, it suggests those fields are missing together — often because they come from the same form section, survey page, or system module that failed or was skipped together.

Q: Why can deleting rows with missing values introduce bias? If the missingness is not completely random (MAR or MNAR), the remaining data after deletion no longer represents the full population, systematically excluding certain types of observations (e.g., non-respondents, high earners) and skewing results.

Q: Suppose a medical dataset has missing test results primarily for low-risk patients. What problems could this create during analysis? The remaining, more complete data would be biased toward higher-risk patients, which could overestimate disease prevalence or severity in the overall population and distort any model trained on this data.

Q: How would you perform EDA on a dataset where multiple columns have different missing-value patterns? Profile missingness per column, visualize patterns with a missingness matrix/heatmap, check correlations among missingness indicators, investigate whether missingness relates to other observed variables

or the target, and decide on a column-by-column (rather than one-size-fits-all) imputation/removal strategy.

Q: How can missing-value analysis reveal differences between customer segments? If certain fields are missing more often for one segment (e.g., new customers vs. long-term customers), that can reflect different onboarding processes, engagement levels, or data collection touchpoints — insight that’s useful beyond just handling the missing values themselves.

5. Outlier Analysis

Q: What is an outlier? An observation that deviates significantly from the rest of the data, either due to genuine natural variability, measurement/data entry error, or a rare but real event.

Q: What are the different types of outliers? Point (global) outliers — a single value far from the rest of the data overall; contextual outliers — unusual only within a specific context (e.g., time or group); and collective outliers — a group of observations that together deviate from the norm, even if individually unremarkable.

Q: Explain the IQR method for detecting outliers. Compute Q1 (25th percentile) and Q3 (75th percentile), then $IQR = Q3 - Q1$. Values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are flagged as outliers. It’s robust to skewed data since it doesn’t rely on the mean/standard deviation.

Q: Explain the Z-score method for detecting outliers. Compute how many standard deviations each value is from the mean: $z = (x - \text{mean}) / \text{std}$. Values with $|z|$ above a threshold (commonly 3) are flagged as outliers. It assumes an approximately normal distribution.

Q: Why can the Z-score method be problematic for heavily skewed data? Both the mean and standard deviation are themselves distorted by skew and extreme values, so the z-score threshold can miss genuine outliers or flag normal values incorrectly.

Q: When would you prefer IQR over Z-score? When the data is skewed, non-normal, or contains extreme values that would distort the mean/standard deviation — IQR gives a more robust and reliable boundary in these cases.

Q: What is the difference between a global outlier and contextual outlier? A global outlier is unusual relative to the entire dataset. A contextual outlier is only unusual within a specific context, such as time of day, season, or subgroup, even though the same value might be normal in another context.

Q: Give an example of a contextual outlier. A temperature of 25°C is normal in summer but would be a contextual outlier if recorded in the middle of winter in the same region.

Q: A person’s electricity consumption is extremely high compared with the overall dataset. How would you determine whether it is an error or a genuine

observation? Check for a plausible explanation (large household, business use, extreme weather, appliance addition), verify against billing/meter records, look for a pattern over time (a one-time spike vs. sustained usage), and consult with domain experts or the data source before deciding.

Q: Why should outliers not automatically be removed? They may represent genuine, important events (fraud, high-value transactions, rare diseases) that are exactly what the analysis or model needs to capture; blind removal can bias results and hide the signal of interest.

Q: What are the different ways to handle outliers? Remove them (if confirmed errors), transform the variable (e.g., log transform) to reduce their influence, cap/winsorize them to a defined boundary, or retain them as-is when they represent genuine and important information.

Q: When would you use winsorization? When you want to reduce the influence of extreme values without discarding the observations entirely — replacing values beyond a chosen percentile with the value at that percentile, preserving sample size while limiting distortion.

Q: What is the difference between removing, transforming, capping, and retaining an outlier? Removing deletes the observation entirely; transforming (e.g., log) changes the scale so extreme values have less leverage; capping (winsorizing) limits extreme values to a defined boundary while keeping the row; retaining keeps the value unchanged because it's considered valid and meaningful.

Q: How can boxplots help identify outliers? Boxplots visually display the median, quartiles, and whiskers (typically at $1.5 \times \text{IQR}$), with individual points beyond the whiskers plotted separately as potential outliers, making them easy to spot at a glance.

Q: A bank transaction dataset contains a few transactions worth ₹10 million while most transactions are below ₹50,000. How would you investigate these observations? Check whether such large transactions are plausible for the account type (business vs. personal), cross-reference with account history and KYC information, check for duplicate or erroneous entries, and consider whether they represent fraud, money laundering flags, or simply legitimate large corporate transfers requiring separate treatment rather than blanket removal.

6. Correlation and Relationship Analysis

Q: What is correlation? A statistical measure of the strength and direction of a linear (or monotonic, depending on the method) relationship between two variables, typically ranging from -1 to +1.

Q: Explain the difference between Pearson, Spearman, and Kendall correlation. Pearson measures the strength of a *linear* relationship between two continuous variables and is sensitive to outliers. Spearman measures the strength of a *monotonic* relationship using ranked values, making it robust to outliers and nonlinearity. Kendall also uses ranks (based on concordant/discordant pairs) and is often more ro-

bust for small samples or data with many tied ranks, though computationally more expensive.

Q: When should Spearman correlation be preferred over Pearson correlation? When the relationship is monotonic but not linear, when the data contains outliers, or when variables are ordinal rather than strictly continuous.

Q: Why does correlation not necessarily imply causation? Two variables can be correlated due to a confounding third variable, reverse causation, coincidence, or shared trend over time, without one directly causing the other.

Q: What is a correlation matrix, and how is it useful during EDA? A table showing pairwise correlation coefficients between all numerical variables in a dataset. It helps quickly identify which variables are strongly related, spot potential multicollinearity, and prioritize which relationships to investigate further.

Q: What is a heatmap, and how can it help identify relationships between variables? A heatmap is a color-coded visual representation of a matrix (often a correlation matrix), where color intensity represents the strength/direction of values, making it easy to visually scan for strong positive or negative relationships across many variable pairs at once.

Q: Two variables have a correlation of 0.95. What conclusions can you safely draw? Only that there is a strong linear (Pearson) association between the two variables in this sample. You cannot conclude causation, that the relationship holds outside the observed range, or that it will hold in other populations/contexts without further investigation.

Q: Two variables have a correlation close to zero. Does that necessarily mean there is no relationship between them? No — correlation coefficients (especially Pearson) only capture linear relationships. Two variables can have a strong nonlinear relationship (e.g., a U-shape) while showing a correlation near zero.

Q: Give an example of a nonlinear relationship where Pearson correlation might be misleading. The relationship between age and physical performance is often U-shaped or inverted U-shaped (performance rises then falls with age), which can produce a Pearson correlation near zero despite a strong underlying relationship.

Q: What is multicollinearity? A situation where two or more predictor variables in a regression model are highly correlated with each other, making it difficult to isolate each variable's individual effect on the target and destabilizing coefficient estimates.

Q: Why is multicollinearity important when performing EDA for regression problems? It can inflate standard errors, make coefficients unstable and hard to interpret, and signal redundant features that might need to be combined, transformed, or removed before modeling.

Q: How can scatter plots reveal relationships that correlation coefficients cannot? Scatter plots visually show the full shape of a relationship — nonlinearity, clusters, heteroscedasticity, or the influence of individual outlier points — none of which are captured by a single correlation number.

Q: How can you investigate whether the relationship between two variables

changes across different groups? Create scatter plots or compute correlations separately for each group/subcategory, or use a grouped/faceted plot (e.g., color-coded scatter plot by group) to visually compare trends across segments.

Q: What is Simpson’s paradox, and why is it important in EDA? Simpson’s paradox occurs when a trend appears in several different groups of data but disappears or reverses when the groups are combined. It’s important because analyzing aggregated data without checking subgroups can lead to entirely wrong conclusions about a relationship.

Q: How can aggregation hide relationships present in the underlying data? Averaging or summing across subgroups can cancel out opposing trends within each subgroup, mask heterogeneity, and produce a misleading overall relationship that doesn’t reflect what’s actually happening within any individual segment.

7. Categorical Data EDA

Q: How should categorical variables be analyzed during EDA? Using frequency counts and proportions, bar charts, checking cardinality (number of unique categories), examining relationships with the target via cross-tabulation or grouped statistics, and checking for rare categories or inconsistent labeling (e.g., “NY” vs. “New York”).

Q: What is the difference between nominal and ordinal variables? Nominal variables are categories with no inherent order (e.g., color, country). Ordinal variables have categories with a meaningful order but no defined numeric distance between them (e.g., “low,” “medium,” “high”).

Q: Why can treating categorical variables as numerical variables lead to incorrect conclusions? Assigning arbitrary numbers to nominal categories implies a false ordering and false mathematical relationships (e.g., that “3” is three times “1”), which can mislead correlation calculations, distance-based algorithms, and regression coefficients.

Q: How can frequency tables help analyze categorical variables? They show the count and proportion of each category, quickly revealing class imbalance, rare categories, unexpected or invalid values, and the overall distribution shape of the categorical variable.

Q: When should you use a bar chart instead of a histogram? Bar charts are used for categorical (discrete, non-ordered/ordered-but-non-continuous) data where each bar represents a distinct category. Histograms are used for continuous numerical data, where bins represent ranges of values along a continuous scale.

Q: A categorical feature has 500 unique values. What EDA questions should you ask before using it? Is this truly categorical or actually an identifier? How is the distribution across categories (are a few dominant, with a long tail of rare ones)? Would grouping rare categories into “Other” help? Would target/frequency encoding be more appropriate than one-hot encoding? Does high cardinality reflect a data quality issue (e.g., inconsistent spelling)?

Q: What problems can high-cardinality categorical variables create? They can cause a dimensionality explosion with one-hot encoding, sparse and unstable estimates for rare categories, overfitting, and increased computational cost, while also making visualizations (like bar charts) cluttered and hard to interpret.

Q: How would you analyze the relationship between customer segment and churn status? Build a cross-tabulation (contingency table) of segment vs. churn, compute churn rate (proportion) per segment, visualize with a grouped/stacked bar chart, and optionally test statistical significance with a chi-square test of independence.

Q: How can cross-tabulation be used during EDA? Cross-tabulation summarizes the joint frequency distribution of two (or more) categorical variables, making it easy to see how categories of one variable relate to categories of another, and to compute conditional proportions and identify associations.

Q: What is the difference between count-based analysis and proportion-based analysis for categorical variables? Count-based analysis looks at raw frequencies, which can be misleading when group sizes differ greatly. Proportion-based analysis normalizes by group size (e.g., churn rate per segment rather than raw churn count), allowing fairer comparisons across groups of different sizes.